TOWER DEFENCE LITE

README — Setup, Controls & Feature Overview

Unity 6 LTS | Android | 3D Tower Defence

Engine	Unity 6 LTS (Universal Render Pipeline)
Platform	Android — min Android 7.0 (API 24)
Scenes	2 — MainMenu and GameScene
Starting HP	20 (set in GameManager Inspector)
Starting Coins	150 + saved coins from previous session
Starting Gems	5 + saved gems from previous session
Target FPS	30+ FPS on 3–4 GB RAM devices
APK size	≤ 50 MB
RAM limit	< 500 MB peak

Setup Instructions

Step 1 — Fix Script Bugs First

Open these files in your code editor and make the changes below before doing anything in Unity:

- WaveData.cs line 3 → change ": MonoBehaviour" to ": ScriptableObject"
- TowerData.cs line 1 → delete the entire line: using UnityEditor.ShaderGraph.Internal;
- ObjectPool.cs → delete: using UnityEngine.InputSystem.Controls;
- EnemyHealthBar.cs → delete: using UnityEditor;

Step 2 — Unity Project

1. Create a new Unity 6 LTS project using the 3D (URP) template.
2. Window → Package Manager → install AI Navigation (for NavMesh / Enemy movement).
3. Window → Package Manager → install TextMeshPro. When prompted, click Import TMP Essentials.
4. Copy all 18 scripts into Assets/Scripts/ keeping the folder structure: Core / Data / Enemy / Tower / Systems / UI.
5. Wait for Unity to compile — Console must show 0 errors before continuing.

Step 3 — Create Both Scenes

- File → New Scene → Save As → Assets/Scenes/MainMenu (exact name)
- File → New Scene → Save As → Assets/Scenes/GameScene (exact name)
- File → Build Settings → Add both scenes. Order: Index 0 = MainMenu, Index 1 = GameScene.

Step 4 — Create ScriptableObject Data Assets

Right-click Assets/Data/ → Create → TowerDefence → EnemyData / TowerData / WaveData.

Asset Name	Type	Key Values to Set in Inspector
BasicEnemyData	EnemyData	enemyName="Basic Enemy" maxHealth=100 speed=3 damage=1 coinReward=10
FastEnemyData	EnemyData	enemyName="Fast Enemy" maxHealth=60 speed=6 damage=2 coinReward=15
BasicTowerData	TowerData	towerName="Basic Tower" damage=20 range=5 fireRate=1 coinCost=50 gemCost=0 lockedByDefault=false
PowerTowerData	TowerData	towerName="Power Tower" damage=40 range=8 fireRate=2 coinCost=0 gemCost=3 lockedByDefault=true
Wave1Data	WaveData	Entry 0: BasicEnemyData count=5 interval=1 timeBetweenWaves=5
Wave2Data	WaveData	Entry 0: BasicEnemy count=5 interval=1 Entry 1: FastEnemy count=3 interval=1 timeBetweenWaves=5

Asset Name	Type	Key Values to Set in Inspector
Wave3Data	WaveData	Entry 0: BasicEnemy count=8 interval=0.8 Entry 1: FastEnemy count=5 interval=0.8 timeBetweenWaves=5

Step 5 — Build Prefabs

Prefab	Base Object	Components to Add	Key Settings
BasicEnemy	Capsule	NavMeshAgent, Enemy.cs, EnemyHealthBar (child canvas+slider)	Layer = "Enemy". poolTag = "BasicEnemy". NavMeshAgent stoppingDistance=0.3
FastEnemy	Capsule	NavMeshAgent, Enemy.cs, EnemyHealthBar	Layer = "Enemy". poolTag = "FastEnemy"
BasicTower	Cube (scale 1×2×1)	Tower.cs	FirePoint child at Y=1.2. projectilePrefab = empty (instant hit)
Projectile	Sphere (scale 0.3)	Projectile.cs	Remove Sphere Collider. speed=10 (default in script)

Step 6 — GameScene Hierarchy Setup

GameObject	Components	Inspector Values
GameManager	GameManager.cs	startingHP=20 startingCoins=150 startingGems=5
WaveManager	WaveManager.cs	Drag Wave1/2/3Data assets. Drag SpawnPoint. Drag Waypoint1–6 in order.
ObjectPool	ObjectPool.cs	Pools: {BasicEnemy, prefab, 10} {FastEnemy, prefab, 8} {Projectile, prefab, 20}
TowerPlacementMgr	TowerPlacementManager.cs	No public fields — nothing to set
UIManager	UIManager.cs	Drag all 17 UI fields (see Design Doc section 6.1)
AudioManager	AudioManager.cs + 2×AudioSource	MusicSource child (loop=true). SFXSource child (loop=false). Add SFX entries: shoot / hit / place / click
Ground (Plane)	Mesh Collider	Scale 8×1×8. Mark Navigation Static. Bake NavMesh.
SpawnPoint	Empty GameObject	X=-35 Y=0 Z=0
Waypoint1–6	Empty GameObjects	See positions in GameScene Setup Guide
TowerSpot1–8	TowerSpot.cs + BoxCollider	Scale 3×0.2×3. Drag MeshRenderer to Spot Renderer field.
PlayerBase (Cube)	—	X=35 Y=1 Z=0 (no script needed)
Canvas	Canvas + CanvasScaler	Scale With Screen Size. Reference: 1080×1920. Match: 0.5

Controls

In-Game (Android touch / Editor mouse)

Action	Input	Script handling it
Select a tower	Tap tower button (bottom bar)	TowerButtonUI.OnClick() → TowerPlacementManager.SelectTower()
Place a tower	Tap a TowerSpot (glowing square)	TowerSpot.OnMouseDown() → TowerPlacementManager.TryPlaceTower()
Pause / Resume	Tap Pause button (top bar)	UIManager.OnPauseButton() — sets Time.timeScale 0/1
Open Shop	Tap Shop button (top bar)	UIManager.OnShopButton() — toggles ShopPanel
Restart game	Tap Restart in any result panel	UIManager.OnRestartButton() — reloads scene
Return to Main Menu	Tap Main Menu button	UIManager.OnMainMenuButton() — LoadScene("MainMenu")
Toggle Music / SFX	Toggle in Settings panel	UIManager.OnMusicToggle / OnSFXToggle → AudioManager

Main Menu Controls

Button	Method	Result
Play	MainMenuUI.OnPlayButton()	SceneManager.LoadScene("GameScene")
Settings	MainMenuUI.OnSettingButton()	Toggles settingPanel.SetActive(!)
Quit	MainMenuUI.OnQuitButton()	Application.Quit()

Feature Overview

Core Gameplay Features

- Enemy pathing — NavMeshAgent follows Transform[] waypoints in sequence. Reaches base → DamageBase(data.damage). Killed → AddCoins(data.coinReward).
- Tower targeting — Physics.OverlapSphere with "Enemy" layer mask. Selects nearest enemy. Instant-hit or projectile attack mode.
- Wave system — WaveManager coroutine. 2s initial delay. Spawns via ObjectPool. Tracks enemiesAlive. Victory when all waves cleared.
- Dual currency — Coins (earned in-game via kills), Gems (starting grant + IAP). Both saved to PlayerPrefs and stacked each session.
- Tower placement — Two-tap flow: select tower type, then tap empty TowerSpot. Cost deducted. Locked towers require shop purchase first.

UI Features

- HUD — Live coin, gem, HP, and wave display. Updated by C# events (OnCoinsChanged, OnHPChanged, OnWaveStart). No per-frame polling.
- Pause menu — Time.timeScale = 0 freezes everything. Resume / Restart / Main Menu buttons.
- Game Over & Victory screens — shows Wave Reached, Enemies Defeated, Coins Earned from GameManager state.
- Shop — ShopUI shows locked towers. Buy with coins or gems. Unlocked status saved permanently via SaveSystem.UnlockTower().
- Message toasts — UIManager.ShowMessage(text) appears 2 seconds then auto-hides. Used for "Not enough currency!" etc.
- Settings — Music and SFX toggles. Saved to PlayerPrefs keys "MusicOn" and "SFXOn".
- Local leaderboard — best wave ever reached, saved to PlayerPrefs "BestWave". Shown in HUD via bestWaveText.

Systems Features

- Object pooling — ObjectPool pre-creates enemies and projectiles. Zero garbage collection during gameplay. Dynamic expansion if pool exhausted.
- Save system — PlayerPrefs wrapper. Saves: coins, gems, best wave, tower unlock flags. All persist between sessions.
- Audio — DontDestroyOnLoad AudioManager. Separate music and SFX AudioSources. PlayOneShot allows overlapping effects.
- ScriptableObjects — EnemyData, TowerData, WaveData. All game values tunable in Inspector without code changes.

Folder Structure

Assets/Scripts/Core/	GameManager.cs
Assets/Scripts/Data/	EnemyData.cs TowerData.cs WaveData.cs
Assets/Scripts/Enemy/	Enemy.cs EnemyHealthBar.cs

Assets/Scripts/Tower/	Tower.cs TowerSpot.cs TowerPlacementManager.cs Projectile.cs
Assets/Scripts/Systems/	WaveManager.cs ObjectPool.cs SaveSystem.cs AudioManager.cs
Assets/Scripts/UI/	UIManager.cs MainMenuUI.cs ShopUI.cs TowerButtonUI.cs
Assets/Prefabs/	Enemies/ Towers/ Projectiles/
Assets/Data/	ScriptableObject .asset files (EnemyData, TowerData, WaveData)
Assets/Scenes/	MainMenu.unity GameScene.unity
Assets/Audio/	Music clips and SFX clips

Tower Defence Lite | README v1.0 | Values verified from script source